

GRADE: 8	NAME: Tales by Dots and Lines
LESSON NO. : 5	
No. of Sessions/Periods	10
Lesson Overview	What is the focus of the Unit? The focus of this unit is to understand the behavior of mean and median with changing data. It introduces the concept of the mean as a "balancing point" and teaches students to interpret and visualize data through line graphs, dot plots, and infographics.
Previous Knowledge	What prior Knowledge the students need in order to be successful with this unit? Students need to recall how to calculate the arithmetic mean (sum of values divided by the count) and find the median (middle value of sorted data) from previous grades.
Vocabulary /Terminology	What is the new vocabulary/ terminology used in this lesson? Arithmetic Mean, Median, Central Tendency, Dot Plot, Frequency Table, Spreadsheet, Cell, Formula, Line Graph, Infographic.
SESSION NO. : 01	CONCEPT NAME: The Balancing Act (Mean as a Center)
Learning Outcomes	Understand the mean as the physical 'center' of a data distribution.
Resources Required – Physical, Digital, Books, Charts, PPT's, Videos, Weblinks, Worksheets etc	Graph paper, counters/beads, .
Teacher Activity	Introduction: Provocation/Recapitulation/Preassessment
Time Assigned: 10 minutes	Provocation by asking students to find the average of 3 and 7, then visualizing it on a number line to show 5 is exactly halfway. Methodology: Use "Math Talk" to explain that the total distances on the LHS of the mean equal the total distances on the RHS
Student Activity	Calculate and mark the mean for collections of 3-4 numbers on dot plots.
Time Assigned: 25 minutes	
Assessment	Verify if the sum of distances on both sides of a proposed mean is equal.
Assignment/HW	Solve "Figure it Out" Q.1: Find the mean of the first 50 natural numbers.
Skills/Competencies	Calculation, Estimation, Logical Reasoning.
Intelligences Addressed	Logical-Mathematical, Visual-Spatial.
SESSION NO. : 02	CONCEPT NAME: Changing Data - Including & Removing Values
Learning Outcomes	Predict how the mean and median change when a new value is added or an existing one removed.
Resources Required – Physical, Digital, Books, Charts,	Whiteboard, dot plot diagrams.

PPT's, Videos, Weblinks, Worksheets etc	
Teacher Activity	Introduction: Provocation/Recapitulation/Preassessment Demonstrate how adding a value greater than the mean increases the mean to maintain balance.
Time Assigned: 10 minutes	Methodology: Comparative analysis of dot plots before and after inserting a new value.
Student Activity	Experiment with removing values to see if the mean stays the same or changes.
Time Assigned: 25 minutes	
Assessment	If a value equal to the mean is added, does the mean change?.
Assignment/HW	Solve "Figure it Out" Q.2: Mark the missing value so the mean remains 9.
Skills/Competencies	Prediction, Pattern Recognition, Analysis.
Intelligences Addressed	Logical-Mathematical, Linguistic (Math Talk).
SESSION NO. : 04	CONCEPT NAME: The Algebra of Mean (Add/Subtract Constant)
Learning Outcomes	Prove how the mean reacts when every data point is shifted by a fixed number.
Resources Required - Physical, Digital, Books, Charts, PPT's, Videos, Weblinks, Worksheets etc	Algebra tiles or digital slides.
Teacher Activity	Introduction: Provocation/Recapitulation/Preassessment Present the algebraic proof showing that adding a constant 'k' to all values adds 'k' to the mean.
Time Assigned: 10 minutes	Methodology: Direct instruction followed by algebraic derivation.
Student Activity	Calculate the mean of {8, 3, 10, 13, 4, 6, 7, 7, 8, 8, 5} and then find the mean when each is increased by 10.
Time Assigned: 25 minutes	
Assessment	Explain using algebra what happens to the average when 2 is subtracted from every value.
Assignment/HW	Solve Q.3 from exercises: Find the correct average height after removing the 1cm shoe error.
Skills/Competencies	Algebraic Thinking, Deductive Reasoning, Accuracy.
Intelligences Addressed	Logical-Mathematical, Verbal-Linguistic.

SESSION NO. : 04	CONCEPT NAME: Multiplying and Doubling Data
Learning Outcomes	Understand the effect of multiplication on the arithmetic mean.

Resources Required – Physical, Digital, Books, Charts, PPT's, Videos, Weblinks, Worksheets etc	Calculators, worksheets.
Teacher Activity Time Assigned: 10 minutes	Introduction: Provocation/Recapitulation/Preassessment Show that if every value in a collection is doubled, the average also doubles. Methodology: Use the distributive property to prove the multiplier effect on the average formula.
Student Activity Time Assigned: 25 minutes	Verify the doubling effect using the data points from the previous session.
Assessment	If every value is multiplied by 5, what is the relationship between the old and new average?.
Assignment/HW	Create two examples of 3 numbers whose mean is 8.
Skills/Competencies	Generalization, Formula Application, Computation
Intelligences Addressed	Logical-Mathematical.
SESSION NO. : 05	CONCEPT NAME: Tinkering with Median
Learning Outcomes	Observe shifts in the median when data points are added.
Resources Required – Physical, Digital, Books, Charts, PPT's, Videos, Weblinks, Worksheets etc	Number cards
Teacher Activity Time Assigned: 10 minutes	Introduction: Provocation/Recapitulation/Preassessment Define median as the middle value where there are equal numbers of values less and greater than it. Methodology: Case study analysis—showing how adding a value greater than the median shifts the center
Student Activity Time Assigned: 25 minutes	Find the median of {8, 10, 19, 92} and determine how adding one value affects it.
Assessment	True/False: "Removing a value less than the median will decrease the median." Justify.
Assignment/HW	Solve Q.9: Determine which values of 'p' keep the median at 29.
Skills/Competencies	Sorting, Categorization, Critical Thinking.
Intelligences Addressed	Logical-Mathematical, Visual-Spatial.
SESSION NO. : 06	CONCEPT NAME: Finding Unknowns & Correcting Errors
Learning Outcomes	Calculate missing values or correct averages using the mean formula.

Resources Required – Physical, Digital, Books, Charts, PPT's, Videos, Weblinks, Worksheets etc	Board/Digital display.
Teacher Activity Time Assigned: 10 minutes	Introduction: Provocation/Recapitulation/Preassessment Work through the "Wrestler Weight" example to find the smudged missing value.
	Methodology: Problem-solving approach using the relationship $\text{Sum} = \text{Mean} \times \text{Count}$.
Student Activity Time Assigned: 25 minutes	Solve the "Coconut Harvest" problem where a 3-unit error affected the average.
Assessment	If the mean of 15 values is 134, find the total sum
Assignment/HW	Find the value of 'y' if the mean of {8, 13, 10, 4, 5, 20, y, 10} is 10.375.
Skills/Competencies	Equation Solving, Data Cleaning, Problem Solving.
Intelligences Addressed	Logical-Mathematical.
SESSION NO. : 07	CONCEPT NAME: Mean and Median with Frequencies
Learning Outcomes	Calculate central tendencies from frequency tables without listing every number.
Resources Required – Physical, Digital, Books, Charts, PPT's, Videos, Weblinks, Worksheets etc	Frequency tables with relevant data
Teacher Activity Time Assigned: 10 minutes	Introduction: Provocation/Recapitulation/Preassessment Explain how to account for the frequency of each value in the average calculation.
	Methodology: Demonstration of "Successive Addition" of frequencies to locate the median position.
Student Activity Time Assigned: 25 minutes	Calculate the average family size for the class using the provided table.
Assessment	Find the median number of times students rode their cycles from the frequency dot plot.
Assignment/HW	Solve Q.11: Find the mean and median for the dart-throwing competition
Skills/Competencies	Tabulation, Large-scale Data Processing, Statistical Analysis.
Intelligences Addressed	Logical-Mathematical, Visual-Spatial.
SESSION NO. : 08	CONCEPT NAME: Using Spreadsheets for Data

Learning Outcomes	Use digital spreadsheets to calculate totals and averages using formulas.
Resources Required – Physical, Digital, Books, Charts, PPT's, Videos, Weblinks, Worksheets etc	Computers/Tablets, Excel or Google Sheets.
Teacher Activity Time Assigned: 10 minutes	Introduction: Provocation/Recapitulation/Preassessment Introduce the structure of cells (A1, B2) and the range expression (Start:End). Methodology: Experiential learning: Writing formulas like =SUM (B3 : G3) and =AVERAGE (B7 : D7) .
Student Activity Time Assigned: 25 minutes	Use the "Grade 8 Marks" table to find total scores for each student.
Assessment	Identify which formula would find the class average in Science.
Assignment/HW	Determine if the class average in Odia is greater than Telugu using the spreadsheet.
Skills/Competencies	ICT Literacy, Data Entry, Computational Thinking.
Intelligences Addressed	Logical-Mathematical, Bodily-Kinesthetic (Typing).
SESSION NO. : 09	CONCEPT NAME: Line Graphs and Trends over Time
Learning Outcomes	Interpret line graphs and compare them with bar graphs
Resources Required – Physical, Digital, Books, Charts, PPT's, Videos, Weblinks, Worksheets etc	Different graphs with relevant data
Teacher Activity Time Assigned: 10 minutes	Introduction: Provocation/Recapitulation/Preassessment Compare clustered-bar graphs with line graphs using Kerala/Punjab temperature data. Methodology: Step-by-step interpretation: Identify (Scale/Labels) then Infer (Patterns/Trends).
Student Activity Time Assigned: 10 minutes	Analyze the "Space Jam" graph to find years with the highest space launches.
Assessment	Identify two consecutive years where the worldwide launch count increased by 2x or more
Assignment/HW	Plot a line graph for the "Shop Customers" visiting vs. purchasing data.
Skills/Competencies	Graphing, Trend Identification, Comparison.
Intelligences Addressed	Visual-Spatial, Logical-Mathematical.

SESSION NO. : 10	CONCEPT NAME: Infographics and Activity Strips
Learning Outcomes	Extract insights from visual infographics and represent personal time-use data
Resources Required - Physical, Digital, Books, Charts, PPT's, Videos, Weblinks, Worksheets etc	"Wheat vs Rice" map, colored paper strips.
Teacher Activity	Introduction: Provocation/Recapitulation/Preassessment Explain how color scales communicate preferences across geographical regions
Time Assigned: 10 minutes	Methodology: Comparison of personal activity strips with family members to find differences in lifestyle.
Student Activity	Use the "Sleep Across Ages" curve to describe the trend from childhood to old age.
Time Assigned: 25 minutes	
Assessment	Based on the map, which are the top 5 states where rice is most popular?.
Assignment/HW	Individual Project: Make an activity strip for different days of the week and calculate average time per activity.
Skills/Competencies	Self-Reflection, Data Visualization, Synthesis.
Intelligences Addressed	Intrapersonal (Self-tracking), Visual-Spatial, Interpersonal (Interpreting others' data).